Geophys. J. Int. (1991) **104**, 319–330

Palaeomagnetism and opaque mineral oxides of some basalt from west central Sinai, Egypt

Nadia A. Wassif
Department of Geophysics, National Research Centre, Dokki-Cairo, Egypt

Accepted 1990 August 23. Received 1990 April 26; in original form 1989 November 10

Downloaded from https://academic.oup.com/gji/article/104/2/319/570616 by Curtin University Library user on 06 May 2026

SUMMARY

A stable remanence was isolated from two intrusive occurrences of volcanic rocks collected from west central Sinai, Egypt. Tertiary basalts at Wadi Nukhul have a predominantly reverse magnetization with mean direction $D = 173.5°$, $I = -60.7°$ ($K = 46.6$, $\alpha_{95} = 8.2°$, dyke and sill) and a palaeopole at 76°S, 193°E. Coarse primary magnetite (100–500 μm) with a Curie temperature between 540° and 580 °C is the principal magnetic mineral.

A Mesozoic basaltic intrusion at Gebel Farsh El Azraq exhibits a predominantly normal magnetization carried mainly by fine-grained (50–300 μm) titanomagnetite. Its mean direction is $D = 139°$, $I = 44.6°$ ($K = 5.9$, $\alpha_{95} = 24.8°$, $N = 8$ sites) with a palaeopole at 28.6°S and 179°E. These results, together with other results from west central Sinai have been compared with the apparent polar wander path (APWP) for Africa given by Irving & Irving (1982). The Mesozoic poles are not compatible with poles of similar age for Africa, reflecting a local effect of tectonic rotations and/or geomagnetic secular variation. In contrast, the Egyptian Early Tertiary poles compare favourably with the relevant part of the African APWP for the period later than 40 Myr, in good agreement with the K–Ar ages of 22–25 Myr obtained for these intrusives.

Key words: Egypt, Oligo–Miocene volcanic event, palaeomagnetism, rock magnetism.

INTRODUCTION

Previous papers (Wassif 1988, 1989) have described the iron–titanium oxide minerals, rock magnetism and palaeomagnetism of some basalts from west central Sinai, Egypt, including Wadi Budra, Wadi Matulla and Wadi Tayiba. The present paper deals with similar studies carried out on two other occurrences, Wadi Nukhul and Gebel Farsh El Azraq. This work includes a comparison of the five palaeopole positions obtained with the apparent polar wander path (APWP) for Africa. The recent work given by Mossa (1987) on eight basaltic occurrences in west central Sinai namely Gebel Farsh El Azraq, Wadi Abu Natash, Wadi Burdra, Wadi Abu Maraq, Wadi Nukhul, Wadi Matulla, Gebel Hammam Faraun and Wadi Tayiba provide a good picture for this area (Fig. 1 and Table 1). Oriented samples were collected from the following sites.

(*a*) *Wadi Nukhul.* This volcanic intrusion (29°N, 33°15′E) is represented by a wide dyke intruding Eocene limestone that contains chert bands. This dyke lies about 8 km SW of Abu Zenima. It is nearly vertical and uniform in width (~7 m). Near its contacts the limestones are baked and show a black colouration. To the east; this dyke extends as a thin sill, conformable with sedimentary beds. Field observations suggest that the Wadi Nukhul intrusion is related to Mid-Tertiary volcanicity. A Miocene age was assigned by Ball (1916) and Hume *et al.* (1920) to this intrusion, while Andrew (1937) and Said (1962) referred it to the Oligocene.

(*b*) *Gebel Farsh El Azraq.* This volcanic intrusion (28°59′N, 33°25′E) is the largest basaltic occurrence in west central Sinai. It is about 30 km SE of Abu-Zenima, at about 790 m above sea-level. It is a thick sill capping Upper Carboniferous sandstones, up to 60 m in thickness and differentiated; a lower sheet (44 m) with a relatively coarse granular texture and an upper one (16 m) with finer texture. The Upper Carboniferous sandstones near the contacts are highly metamorphosed to meta quartzites. Barron (1907) and Barthoux (1922) regarded the basalts of Gebel Farsh El Azraq as of Carboniferous age, while Andrew (1937) considered them to be of the same age as the other Late

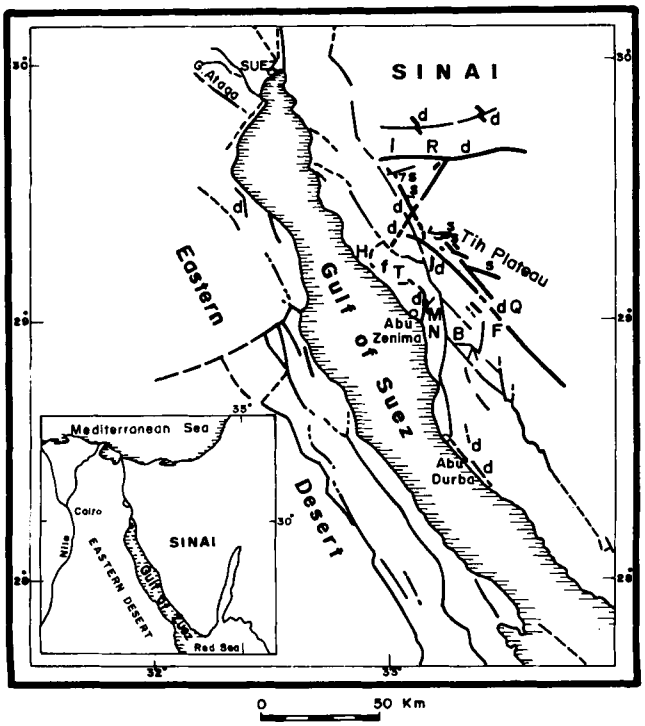


Figure 1. Sketch map showing the distribution of the different igneous bodies, west central Sinai, Egypt. —; igneous body, A; Wadi Abu Natash, N; Wadi Nukhul, —; major fault, B; Wadi Budra, Q; Wadi Abu Maraq, d; Dike, F; Gebel Farsh El Azraq, R; Raqabat e-Na'ama, S; Sill, H; Gebel Hammam Faraun, T; Wadi Tayiba, f; Flow, M; Wadi Matulla.

Tertiary basalts. El-Shazly *et al.* (1974), regarded the Frash El Azraq basalts as volcanics of Jurassic age.

SAMPLING AND MEASUREMENT PROCEDURES

Forty oriented basaltic hand samples were collected from eight sites covering the two sheets of Gebel Farsh El Azraq and 25 hand samples were obtained from the dyke and sill of Wadi Nukhul. The samples were oriented by magnetic compass in the field and cored in the laboratory. Cores were sliced into one to three specimens (2.5 cm diameter, and 2.5 cm height) and their natural remanent magnetization (NRM) was measured with a digital Schonstedt DSM-1 spinner magnetometer. Usually two specimens per sample were stepwise demagnetized in alternating fields (AF), in 12 steps up to 100 mT using a Schonstedt GSD-1 demagnetizer. The residual static field in the shield is less than 10 nT. Unreliable measurements were repeated or discarded. Thermomagnetic curves were measured in air for 10 representative samples from the two occurrences using a recording Curie balance in an applied field of 3.1 A m^{-1}. Initial susceptibility was measured by an inductometer. The mineralogy was investigated by reflected-light microscopy (one polished surface from each hand sample).

OPAQUE PETROLOGY

Petrographic studies show that the volcanics of Gebel Farsh El Azraq are olivine basalts, while the basaltic rocks at Wadi Nukhul (dyke) are mainly olivine dolerities with subophitic and intergranular textures. The opaque minerals present and their distribution, grain sizes, textures and degree of alteration and oxidation were determined (Table 2). A characteristic feature of the Wadi Nukhul samples is the apparent absence of exsolved ilmenite lamellae in magnetite. These samples are rich in primary sulphides; this may be due to their intrusive character, as sulphur normally escapes with volcanic gasses from extrusive igneous rocks. In contrast, samples from Gebel Farsh El Azraq (lower sheet) show a very advanced alteration of the titanomagnetite which includes the formation of pseudobrookite.

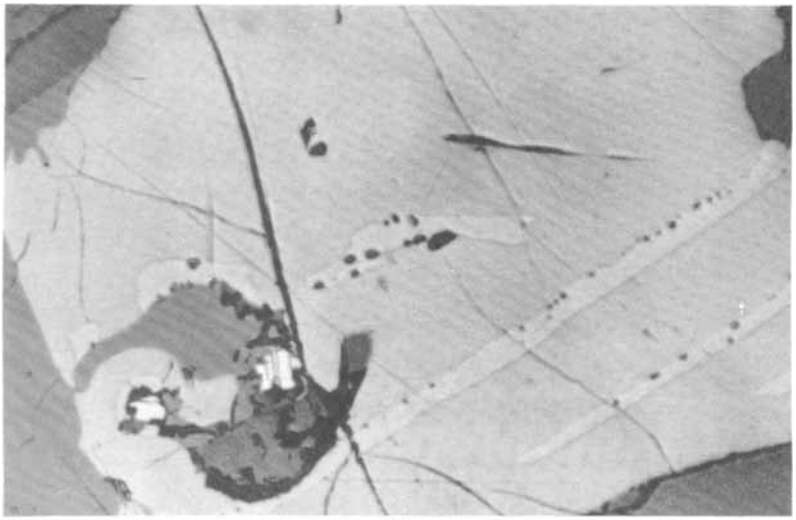
THERMOMAGNETIC MEASUREMENTS

Six determinations of Curie temperatures were made on samples from each occurrence.

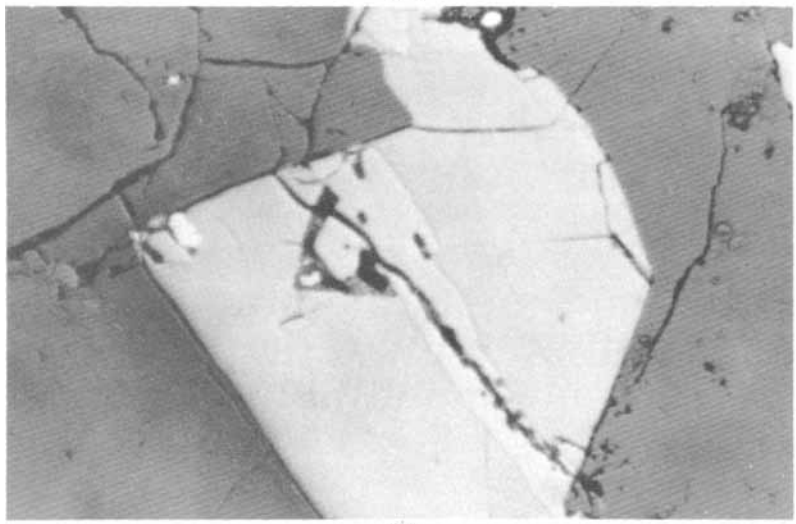
Table 1. Locality description.

Geological setting (Mossa, 1987)	Petrology	Principal magnetic minerals
Gebel Farsh El Azraq A thick sill capping Upper Carboniferous sandstone reaching about 60m thickness which can be divided into 2 units.	Olivine basalt	Mainly homogeneous titanomagnetite
Wadi Abu Natash Several tabular bodies occur along the northern and southern side of Wadi Abu Natash, overlying the Lower Corboniferous sandstones related to the Ataqa Formation	"	"
Wadi Budra A lava flow in the form of a basaltic sheet separating the underlying Ataqa Formation from the overlying Budra Formation.	"	"
Wadi Abu Maraq- Wadi Nukhul -Wadi Matulla Doleritic dykes cut Eocene rocks but never intrude the Early Miocene sediments. They vary greatly in thickness, extension and direction with the predominance direction parallel to the Gulf of Suez and Gulf of Aqaba.	Olivine dolerites with subophitic and intergranular textures	Titaniferous variety of magnetite showing different fabrics of ilmenite intergrowths
Gebel Hammam Faraun A thick doleritic sill that intrudes the rocks of the main western scarp of Gebel Hammam Faraun. The underlying and overlying sediments are most probably of Cenomanian and Turonian ages respctively.	"	"
Wadi Tayiba A lava flow which is underlain by Upper Eocene red beds and overlain by basal Miocene rocks.	Porphyritic olivine basalt	"

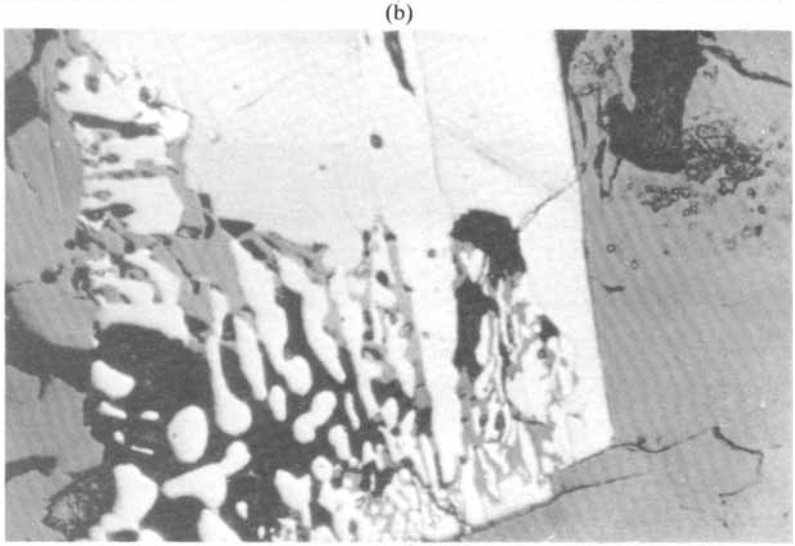
Downloaded from https://academic.oup.com/gji/article/104/2/319/570616 by Curtin University Library user on 06 May 2026



(a)



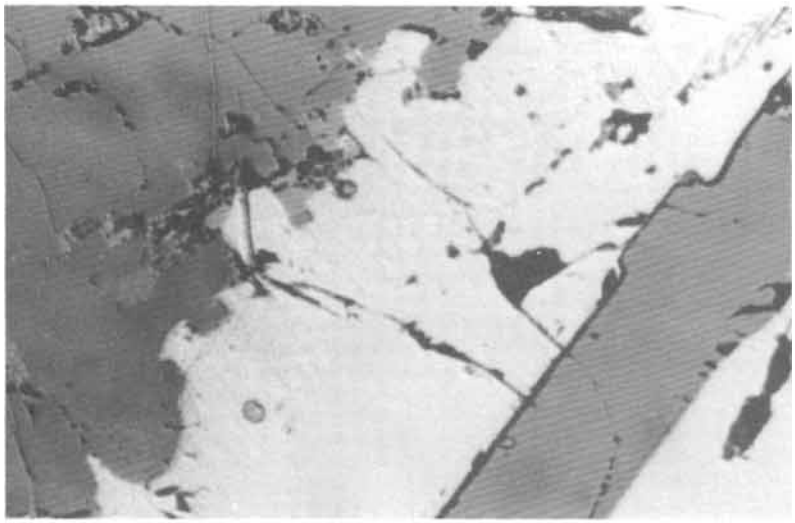
(b)



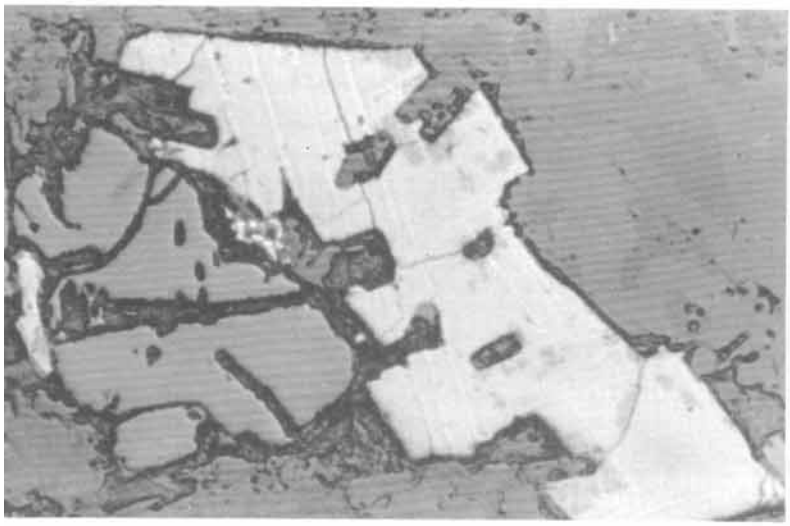
(c)

Figure 2. (a) Magnetite (coarse grained) containing irregularly distributed broad lamellae of ilmenite (trellis texture). Notice the sulphide granules in the silicates. Nukhul dyke. Reflected light, ×100. (b) Sandwich intergrowth showing one broad band of ilmenite in a host homogeneous magnetite, Nukhul dyke. Reflected light, ×100. (c) Graphic or myrmikitic intergrowths of magnetite with secondary amphibole; the magnetite also contains a sandwich intergrowth of ilmenite. The latter have tapering ends and penetrate the myrmikitic magnetite intergrowth. Nukhul dyke. Reflected light, × 100. (d) Ilmenite and magnetite in juxtaposition. The magnetite contains fine exsolved lamellae of ilmenite and also shows signs of martitization at its contact with the ilmenite lamellae. Upper sheet, Gebel Fars El Azraq. Reflected light, ×100. (e) Pseudobrookite (dark grey) forming irregular patches in a titanomagnetite grain. The latter contains coarse trellis lamellae of ilmenite. Lower sheet, Farsh El Azraq. Reflected light, ×100. (f) More advanced replacement of titanomagnetite by pseudobrookite. Lower sheet, Farsh El Azraq. Reflected light, ×100.

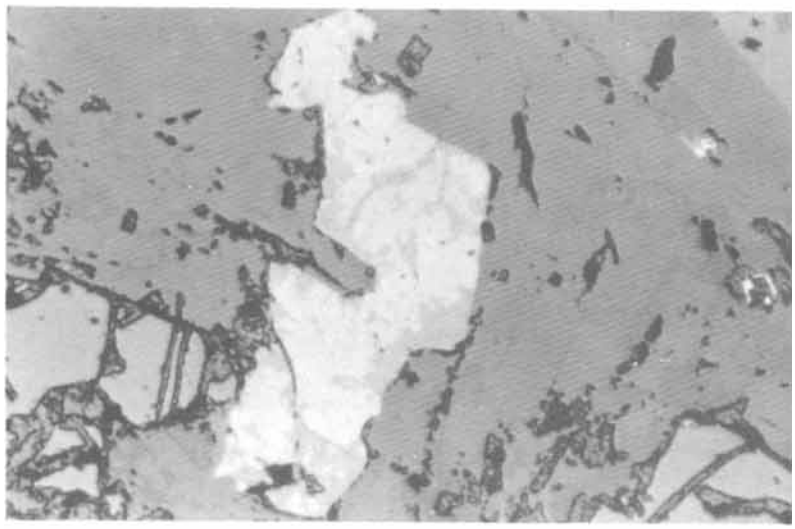
Downloaded from https://academic.oup.com/gji/article/104/2/319/570616 by Curtin University Library user on 06 May 2026



(d)



(e)



(f)

Figure 2. (*continued*)

Downloaded from https://academic.oup.com/gji/article/104/2/319/570616 by Curtin University Library user on 06 May 2026

Table 2. Ore microscopic results.

Locality	Opaque minerals		Magnetite		Ilmenite	Sulphides
	% by volume	percentage and grain size	Type and intergrowths	Alteration		
Wadi Nukhul Dyke	8-10	60-70% coarse grained (100-500 μm) - 80 %	Titaniferous magnetite[a] showing different fabrics of ilmenite intergrowths Trellis texture (Fig. 2a) Sandwich texture (Fig. 2b) Myrmikitic texture (Fig. 2c)	Remarkably fresh	- 30-40% homogeneous elongated plates (150-300 μm in length) - 20 %	1-2% Numerous (15-50 μm)
Sill	10	Very fine grained (up to 50 μm)	Minute grains of magnetite indicating rapid cooling		Homogeneous minute laths	
Gebel Farsh El Azraq		- 70 %	Mainly titanomagnetite[a] and lesser amounts of titaniferous magnetite. Intergrowths of sandwiches and composite grains in juxtaposition (Fig. 2d)	Normal martitization	- 30 % Homogeneous thin plates and needles (up to 150μm in length).	< 1 %
Upper sheet (16 m)	3-5	Fine to medium grained (50-200 μm)				
Lower sheet (44 m)	3-5	- 70 % Medium grained (100-300 μm)	Mainly or only titanomagnetite[a] Few trellis textures at the to pmost part of the layer (Fig. 2 e)	Dark grey pseudobrookite (in the top) due to the superheating caused by the intrusion of the upper sheet (Fig. 2 f)	- 30 % Homogeneous plates terminated by rhombohedral faces (up to 250 μm in length)	< 1 %

[a] The term titanomagnetite indicates a homogeneous single-phased magnetite containing Ti in solid solution (Basta, 1970) While the term titaniferous magnetite is used for magnetite containing different varieties of ilmenite , exhibiting various textures .

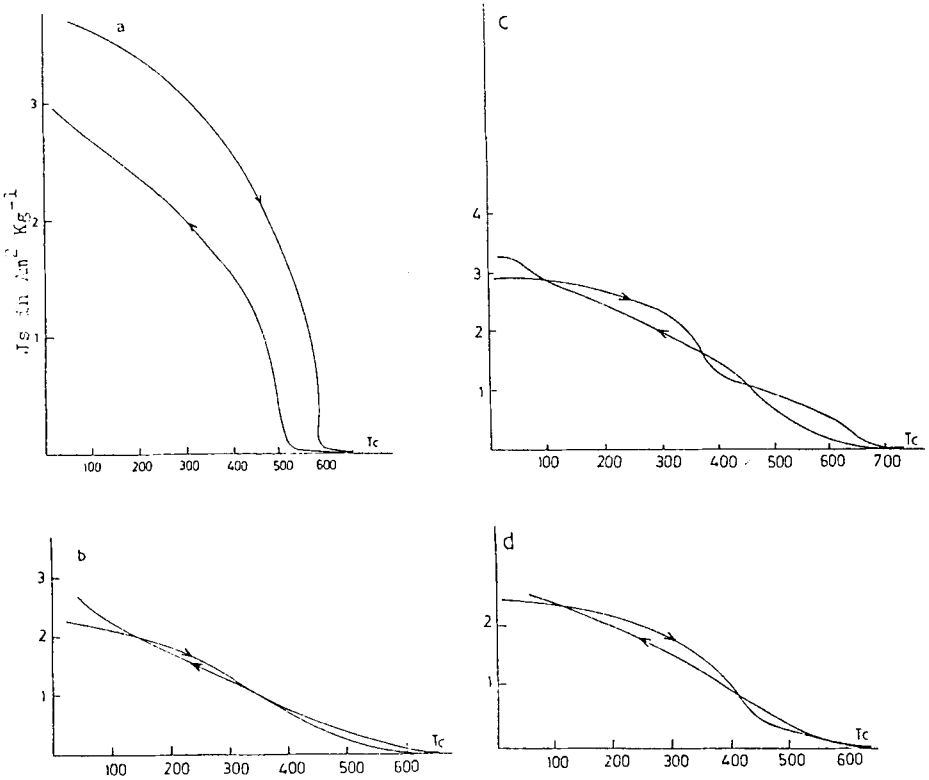


Figure 3. Representative J_s–T curves for samples from each occurrence. Both heating and cooling cycles are shown and indicated by arrows. Note that saturation magnetization values are given in A m^2 kg^{-1}.

Downloaded from https://academic.oup.com/gji/article/104/2/319/570616 by Curtin University Library user on 06 May 2026

Wadi Nukhul. The thermomagnetic curves were fairly uniform (Fig. 3a), being essentially irreversible with heating and cooling, and showed a single well-defined Curie point. The Curie points of all the representative samples fall in the range 540°–580 °C, i.e. near to that of magnetite. The Curie point on the cooling portion of the curve (Fig. 3a) has a lower value (typically about 520 °C) related to a high titanium content accompanied by an observable decrease in magnetization after thermal cooling.

Gebel Farsh El Azraq. The J_s–T curves (Figs 3b, c and d) were mostly irreversible with the cooling curve being slightly lower than the heating curve, although a few samples (lower sheet) showed a more rapid initial decrease in magnetization (Fig. 3b) probably due to alteration (see Table 2). All the samples show two distinct Curie points (Figs 3b, c and d), the first at 350°–415 °C with clear inflections, and the second at 600°–665 °C. The lower Curie points reveal that the magnetic mineral present is titanomagnetite (Ozima & Larson 1970). Ore microscopy supports this interpretation.

DEMAGNETIZATION ANALYSIS

In general, straight line segments on orthogonal projections of demagnetization data define single components of natural remanent magnetization (NRM). Although it is often difficult to determine which points on a demagnetization trajectory 'belong' to a straight line segment, stable endpoint magnetization is accepted provided it can be experimentally confirmed by a minimum of three successive demagnetization steps (per specimen) associated with a steady intensity reduction.

Wadi Nukhul. NRM directions are grouped in most specimens, with a predominance of steep to intermediate negative inclinations. NRM intensities are variable, ranging from 0.63×10^{-3} to 0.24×10^{-1} A m^{-1}. On AF demagnetization, most specimens reach a stable endpoint at 35 mT (specimens 11N, 12N and 16N in Fig. 4), while others reach this at only 20 mT (specimens 13N, 14N and 15N). This has a south southeast declination with a steep negative inclination (Figs 4a–f). The median destructive field values MDF (the magnitudes of the peak alternating field required to reduce the NRM intensity to one half of its initial value) range from 10 to 40 mT. This parameter is a measure of the stability of remanent magnetization. Some specimens (e.g. 12N and 16N, Figs 4d and f) showed substantial changes in direction, their intensities also increasing with increasing demagnetizing field up to nearly 30 mT, then dropping abruptly at 40 mT. This may indicate the presence of a relatively large secondary component (85 per cent of the initial value) superimposed on the primary component of magnetization. For sill specimens, the principal directions of magnetization do not change significantly with demagnetization up to 100 mT and the remanent intensities decrease gradually, but do not reach the origin (Fig. 4e). This behaviour may be due to the very fine grain size nature of the sill which indicates rapid cooling.

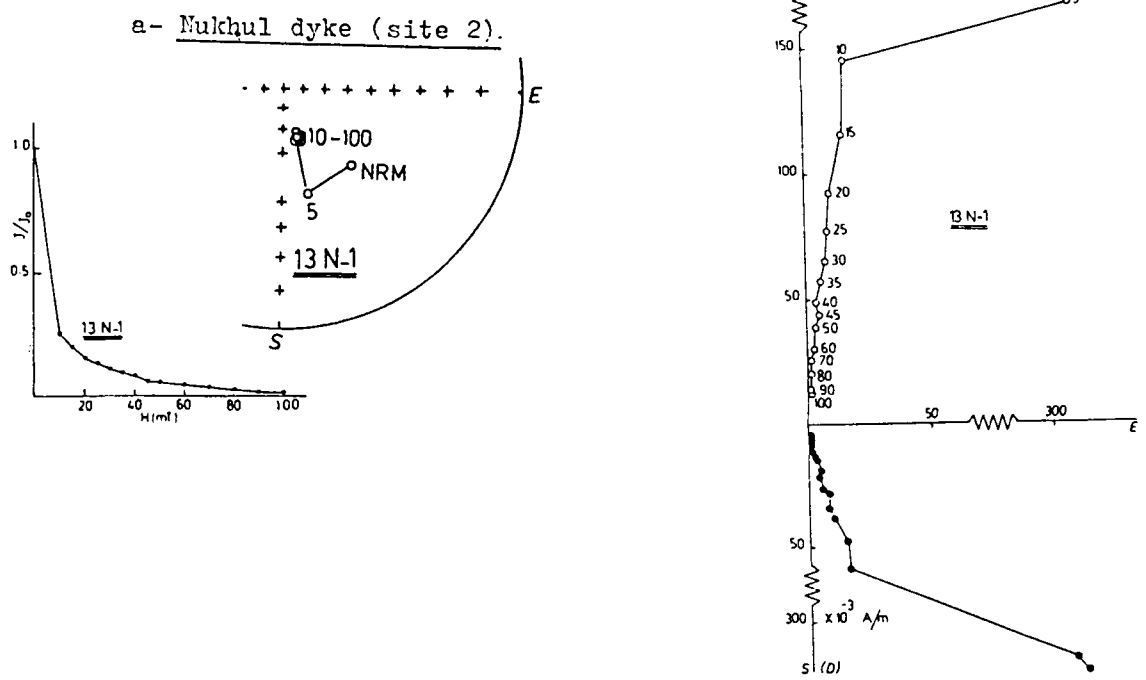


Figure 4. Directions and intensities obtained by AF demagnetization of specimens from Wadi Nukhul. Intensities (J) are normalized to the NRM (J_0) value. Directions are plotted on an equal-area projection with open (closed) symbols indicating upward (downward) vectors. Zijderveld diagrams (Zijderveld 1967) show the projection of magnetization vectors onto the horizontal plane (closed symbols) and north–south vertical plane (open symbols). Alternating field values are in mT.

Downloaded from https://academic.oup.com/gji/article/104/2/319/570616 by Curtin University Library user on 06 May 2026

Gebel Farsh El Azraq. Magnetization directions for most of the specimens from the two sheets show some changes during demagnetization and a preference toward the SE gradient with a positive inclination. NRM intensities are low, ranging from 0.13×10^{-3} to 0.13×10^{-1} A m^{-1}. Four representative examples of orthogonal projections of demagnetization trajectories are shown in Fig. 5(a–d). All specimens exhibit a stable remanent component after demagnetization at 15 mT. There is a decrease to one half and less (30–50 per cent) of the initial magnetic intensity values after eliminating lower coercivity components, between 10 and 15 mT. The results show some differences between the two sheets. The direction for the lower sheet has a S–E declination, intermediate inclination and the remanent magnetization reaches the origin after treatment at 100 mT (Figs 5a and b). The directions for the upper sheet are much steeper, to the S–E and the remanent magnetization reaches the origin about 60 mT (Figs 5c and d). The intensity decay curves show a rapid decrease indicating low coercivities (magnetic softness) where the MDF values range from 10 to 15 mT.

PALAEOMAGNETIC RESULTS

The mean directions (Fisher 1953) of stable remanent magnetization for all sampling sites, assigning unit weight to

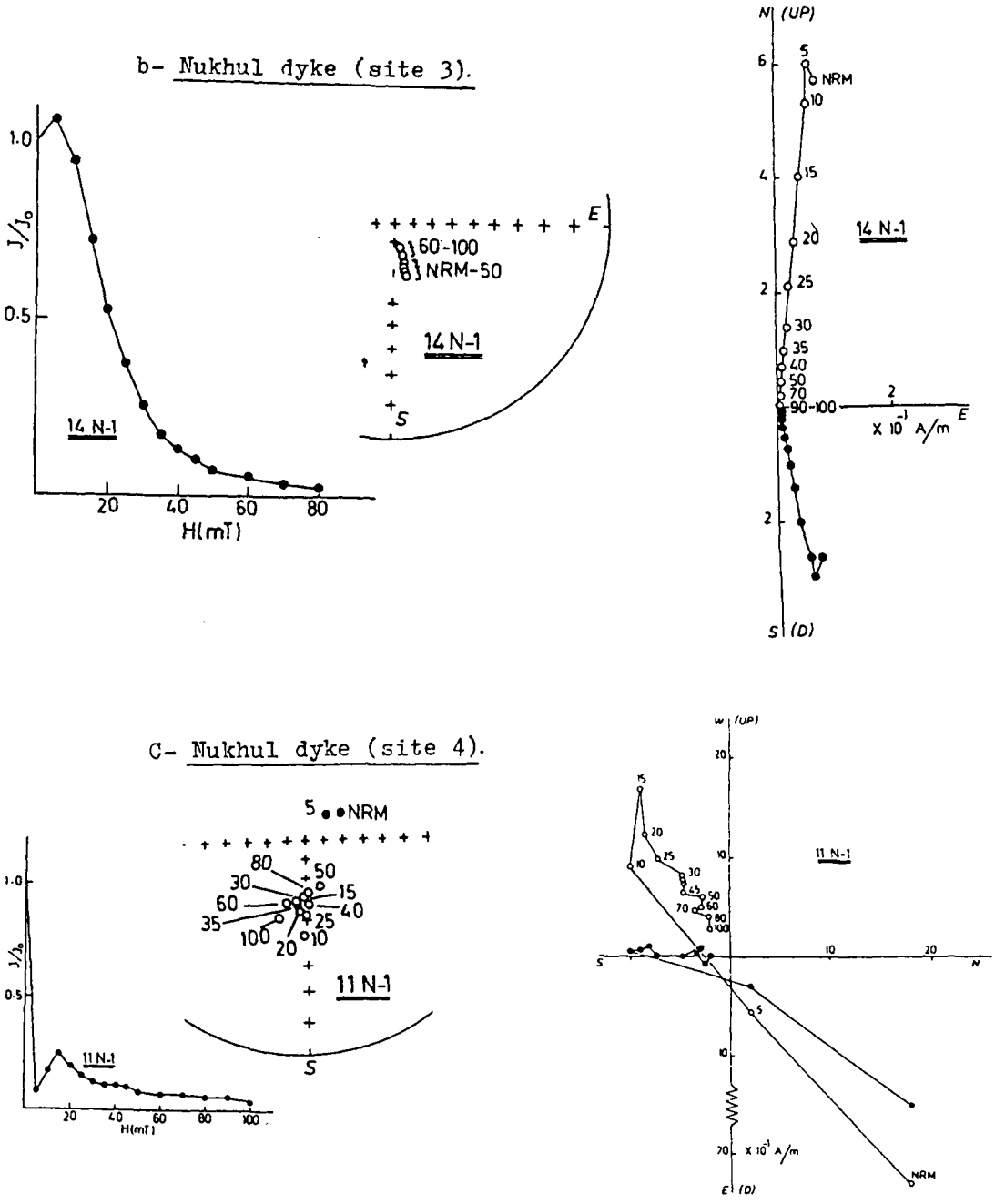


Figure 4. (*continued*)

Downloaded from https://academic.oup.com/gji/article/104/2/319/570616 by Curtin University Library user on 06 May 2026

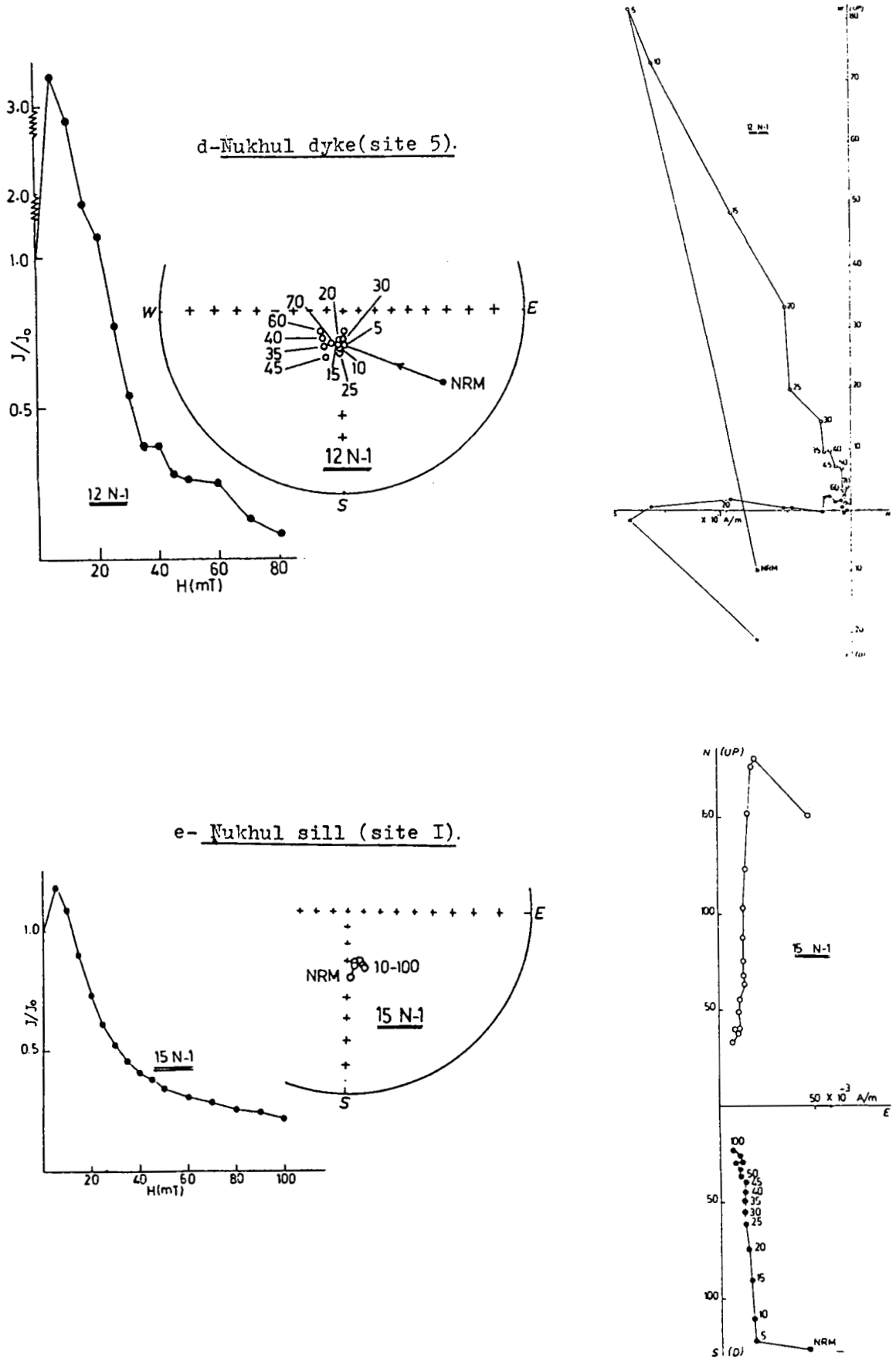


Figure 4. (*continued*)

Downloaded from https://academic.oup.com/gji/article/104/2/319/570616 by Curtin University Library user on 06 May 2026

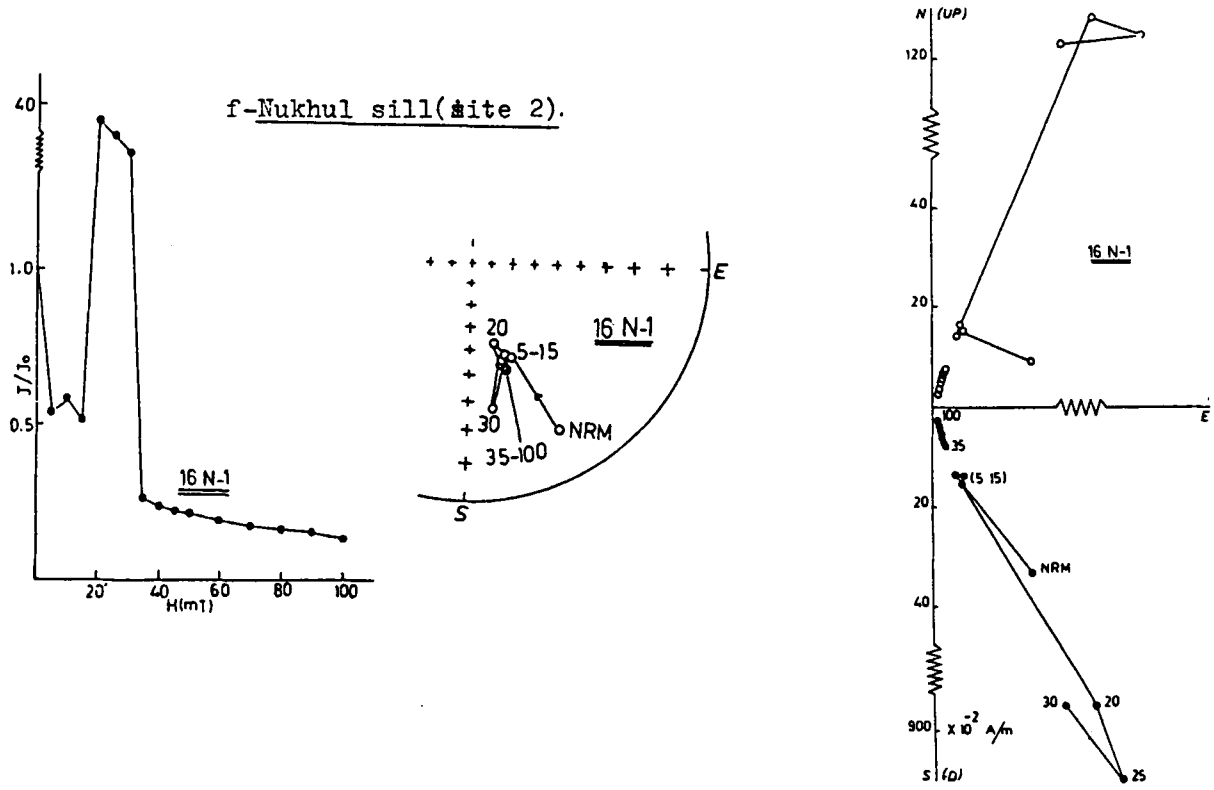


Figure 4. (*continued*)

the mean direction for each hand sample are shown in Fig. 6(a) and listed in Table 3. The specimens are all reversely magnetized and well grouped in the south–southeast, with a mean direction of $D = 173.7°$, $I = -60.7°$ ($K = 46.6$, $\alpha_{95} = 8.2°$).

The site-mean directions for the upper and lower sheets of Gebel Farsh El Azraq (Table 4) are presented in Fig. 6(b). All the eight sites have a normal polarity (i.e. positive inclination). The results are rather scattered in the south–east gradient with a mean direction of $D = 139°$, $I = 44.6°$ ($K = 5.9$, $\alpha_{95} = 24.8°$). The palaeomagnetic poles determined for Wadi Nukhul and Gebel Farsh el Azraq are 79°S, 193°E ($dp = 9.5$, $dm = 12.5$) and 28.6°S, 179°E ($dp = 14.2$, $dm = 22.5$) respectively.

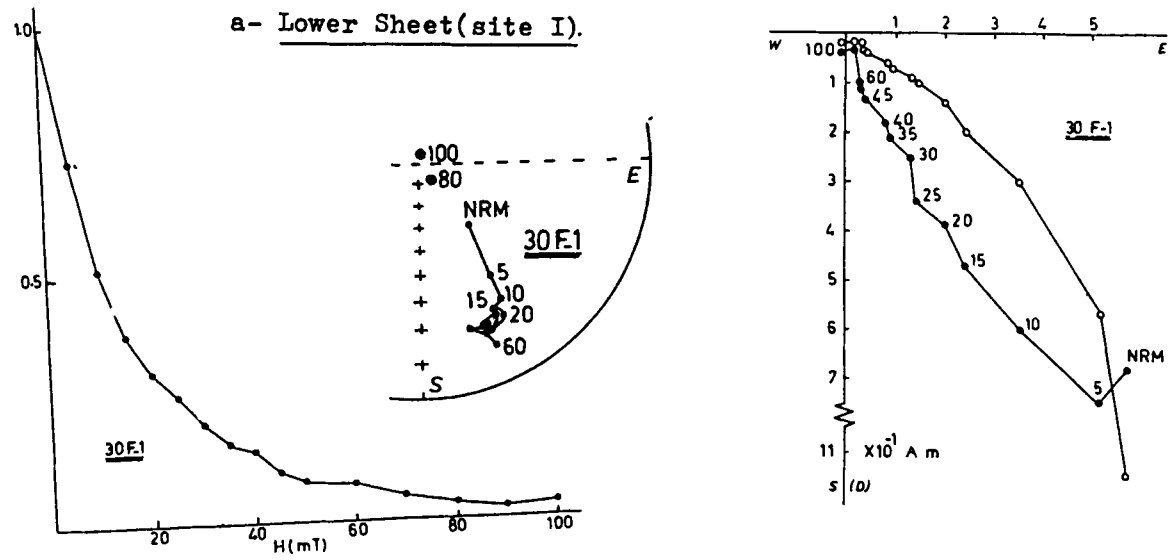


Figure 5. Some representative examples of AF demagnetization results for specimens from Gebel Farsh El Azraq. Symbols as in Fig. 4.

Downloaded from https://academic.oup.com/gji/article/104/2/319/570616 by Curtin University Library user on 06 May 2026

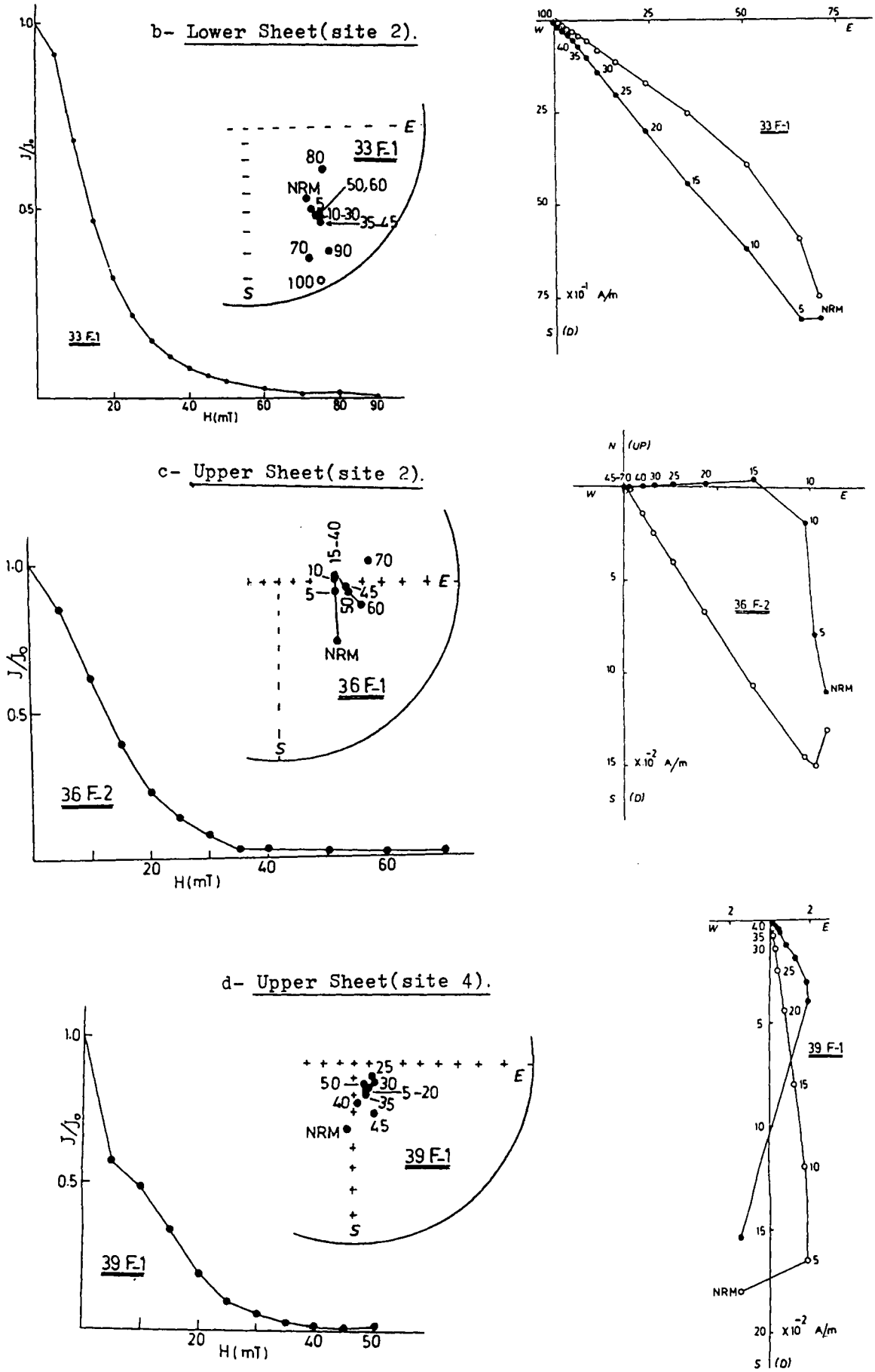


Figure 5. (*continued*)

Downloaded from https://academic.oup.com/gji/article/104/2/319/570616 by Curtin University Library user on 06 May 2026

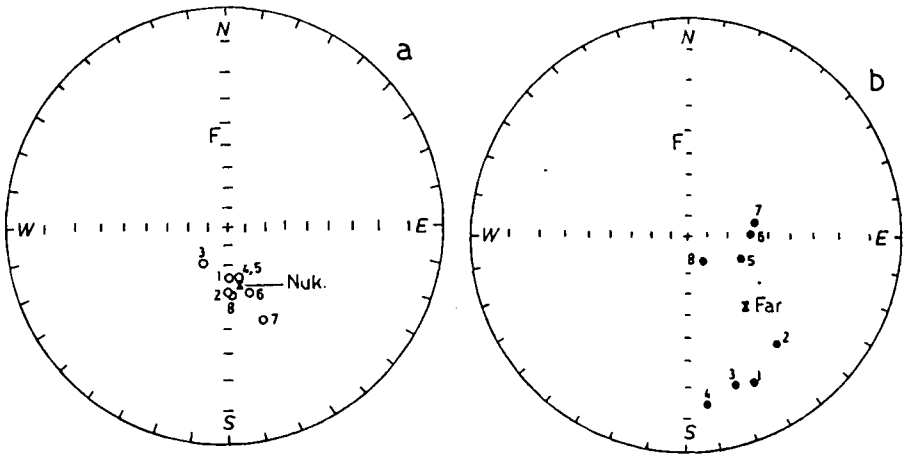


Figure 6. Stereoplot showing the site mean directions of magnetization after optimum AF cleaning. Open (full) symbols = negative (positive) inclinations. F = present field direction, Nuk = the mean for Wadi Nukhul and Far the mean for Gebel Farsh El Azraq.

Table 3. Palaeomagnetic directions and pole positions for Wadi Nukhul.

Site	N	D°	I°	K	α_{95}°	Site lat. (N°)	Site long (E°)	Palaeomagnetic pole lat. (S°)	Palaeomagnetic pole long.(E°)	dp	dm
Dyke											
1	3	177	-65.33	100	12.4						
2*	2	179	-57.5	-	-						
3*	2	212.9	-68.4	-	-	29°	33° 25'				
4	2	167.7	-64.8	-	-						
5	3	167.0	-64.4	66.6	15						
Mean	5	178.8	-64.8	80	8.6			-72	210	11	13.8
Sill											
6	5	163.4	-56.3	80	8.6						
7	4	159.8	-41.6	75	10.7						
8	4	179.4	-57.5	44	16						
Mean	3	166.6	-52	50	17.6			-77.9	137	9.6	24
Combined rock unit		173.7	-60.7	46.6	8.2			-79	193	9.5	12.5

Notes: N is the number of samples; D & I are declination and inclination. K is Fisher's precision parameter. α_{95}° is the semi-angle of the cone of confidence.
* Statistical parameters not given for n < 3.

Table 4. Palaeomagnetic directions and pole positions for Gebel Farsh El Azraq.

Site	N	D°	I°	K	α_{95}°	lat. (N°)	long (E°)	Palaeomagnetic pole lat. (S°)	Palaeomagnetic pole long.(E°)	dp	dm
a) Lower-sheet											
1	4	155	17.9	60	11.9						
2	5	139	25	133	6.6						
3	6	160.8	18.9	62.5	8.5	28° 59'	33° 25'				
4	5	173.6	15.4	200	5.4						
Mean	4	157	19.7	30	17			-47.8	188	9.3	17.7
b) Upper sheet											
5	5	112	60	100	7.6						
6	6	88.9	56.4	166.6	5.2						
7	4	78.6	54	42.8	14.2						
8	4	147.3	74.6	60	11.9						
Mean	4	99	63	27	17.9			2.4	78	22	28
Whole Rock unit	8	139	44.6	5.9	24.8			-28.6	179	14.2	22.5

Symbols are the same as in Table 3

Downloaded from https://academic.oup.com/gji/article/104/2/319/570616 by Curtin University Library user on 06 May 2026

DISCUSSION AND CONCLUSIONS

A directionally stable component of NRM was isolated by AF demagnetization in individual samples from the two igneous bodies studied. Median destructive field values were relatively high for the Nukhul basalts (15–40 mT) but rather low (~10 mT) for the Gebel Farsh El Azraq rocks. The Koenigsberger ratio $Q_n = J_{NRM}/XF$ is a measure of the relative importance of remanent to induced magnetization in a sample where J_{NRM} is the intensity of NRM. X is the initial susceptibility and F is the geomagnetic field intensity (Koenigsberger 1938) at the sampling locality. The Q_n ratios for the Nukhul samples ranged from 4.09 to 16.45 and those for the Farsh El Azraq samples from 2.62 to 10.10. Therefore, remanent magnetization dominates the total magnetization for these basalts in the Earth's magnetic field. However, generally speaking the MDF values correlate with Q_n (e.g. Dunlop & Prevot 1982) and the observed values probably indicate variable stability of the NRM. One of the main differences between the two occurrences is that samples from Gebel Farsh El Azraq have an arithmetic mean NRM intensity (J_{NRM}) value of 4.6×10^{-3} A m^{-1} and a Q_n value of 4.2 while the Nukhul samples have $J_{NRM} = 15.3 \times 10^{-3}$ A m^{-1} and a correspondingly high $Q_n = 9.5$. The high intensities of the Wadi Nukhul basalts probably result from the presence of magnetite as the main magnetic mineral and the abundance of the opaque oxide minerals. The presence of magnetite is also suggested by the observed Curie Point values (540°–580 °C) for this locality. Under the microscope, polished sections show titaniferous magnetite with different fabrics of ilmenite intergrowth (see Table 2). The sill remanence is characterized by higher stability to alternating field demagnetization, and is carried by unaltered very fine grained magnetite. The Gebel Farsh El Azraq samples reveal low Curie temperatures (350°–415 °C) which indicate the presence of titanomagnetite as indicated by the low coercive forces deduced from the AF demagnetization. Ore microscopy supports this conclusion. The low intensity values for this unit are possibly due to a combination of the effects of alteration and the low percentage of opaque minerals (see Table 2).

Mesozoic palaeomagnetic poles

Little igneous activity occurred in the Gulf of Suez region after the Precambrian. A few basaltic bodies of Late Jurassic or Cretaceous age occur west of the Gulf (Abdallah, El-Adinani & Fahmi 1963; Meneisy & Kreuzer 1974). The isotopic age determinations obtained for the Gebel Farsh El Azraq basaltic rocks yielded a mean value of 243 ± 8 Myr (Mossa 1987) which is in good agreement with that given by M. Y. Meneisy (private communication) derived from two size fractions of the same sample from Gebel Farsh El Azraq, and yielding age values of 237.9 ± 3 and 232.9 ± 3 Myr. The basalt occurrence at Wadi Budra which separates the underlying Ataqa Formation from the overlying Budra Formation, yielded a K–Ar age of 182 ± 7 Myr (Mossa 1987). The basaltic sill at Wadi Abu Natash that also caps the Ataqa Formation and which is the northeastern continuation of the Wadi Budra sill has been dated by Weissbrod (1969) at 178 ± 12 Myr. Thus the basaltic ocurrences at Gebel Farsh El Azraq on the one hand and those at Wadi Budra and Wadi Abu Natash on the other, represent two distinct episodes of volcanic activity in west central Sinai, which occurred in Middle Triassic and Middle Jurassic times respectively.

The palaeopole positions obtained for Farsh El Azraq and Wadi Budra (Wassif 1988, 1989) are given in Table 5 and shown in Fig. 7. The K–Ar ages of the associated occurrences mentioned by Mossa (1987) are given in the same table and are included in the following discussion. For the evaluation of the present data on an African scale, these results are compared with the review of palaeomagnetic results given by Irving & Irving (1982) which provides a good picture of the African APWP from the Carboniferous to Recent. This comparison shows that the Middle-Triassic and Middle-Jurassic K–Ar age determinations are not in accordance with the palaeomagnetic results from Gebel Farsh El Azraq and Wadi Budra as these are offset from other Triassic–Jurassic African poles (Fig. 7). This could reflect local tectonic rotations of this part of the African plate relative to the rest of Africa. The tectonic setting (Mossa 1987) indicates that the older Triassic–Jurassic basalts are subaerial fissure eruptions and the continental magmatic activity is possibly connected with the Mesozoic break up of Gondwanaland.

Initial rifting of Pangaea most probably occurred during the Triassic (Dietz & Holden 1970). The fit of eastern and western Gondwanaland has been, and still continues to be, in dispute (Embleton & McElhinny 1975; McElhinny & Embleton 1976; Tarling & Kent 1976; Tarling 1980). However, the individual palaeomagnetic directions for both

Table 5. Summary of west central Sinai palaeomagnetic and pole position results.

	Position of rock unit Lat. (N°)	Long. (E°)	I°	D°	K	α_{95}°	θ_{63}°	δ°	Age (Mossa 1987)	pole position (S°)	(E°)
Farsh El Azraq	28 59	33 25	+45	139	6	25	33	32	243 ±8 (Middle Triassic)	29	179
Wadi Budra	28 57	33 19	+40	159	9	26	27	26	182± 7Ma (Middle Jurassic)	35	190
Wadi Tayiba	29 12	33 05	-10	188	267	2	5	4	22.8 ±1.4 Ma	65	238
Wadi Matulla	29 03	33 10	-56	194	272	6	5	4	23.7 ±1.9 Ma	76	269
Wadi Nukhul	29 -	33 15	-61	174	80	9	9	11	24.8 ±1.6 Ma	79	193

Downloaded from https://academic.oup.com/gji/article/104/2/319/570616 by Curtin University Library user on 06 May 2026

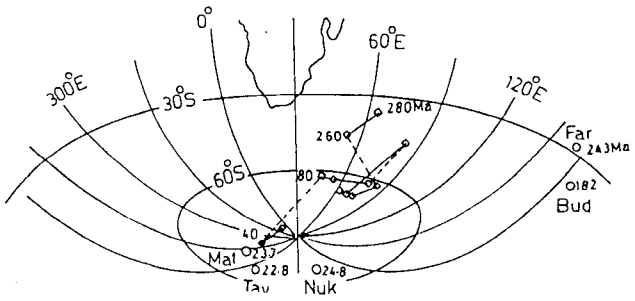


Figure 7. Mean palaeopoles for Africa for the period Carboniferous to Recent from Irving & Irving (1982). The calculated mean palaeopoles (diamonds) are shown at 20 Myr intervals by averaging the available data over a moving window of 30 Myr duration. Means are omitted if only one palaeopole appeared in the window. The data for Sinai are shown as circles: Far = Farsh El Azraq, Bud = Budra, Tay = Tayiba, Mat = Matulla and Nuk = Nukhul. Corresponding ages (in Myr) are shown.

the Farsh El Azraq and the Budra localities may have limited significance because each occurrence is a single cooling unit which cannot have averaged geomagnetic secular variations. Furthermore the observed small values of the precision parameter K, and the large values of the semi-angle of the cone of confidence α_{95}, and the circular standard deviation θ_{63} (Fisher 1953) indicate a large within-site scatter in the directions for these two localities (Table 5). The circular standard deviation values are scattered and fall in the range 5.7°–33.3° for the Farsh El Azraq sites and 27.3°–40.5° for the Budra sites.

Tertiary palaeomagnetic poles

Numerous lava flows, dikes and sills are known in Sinai near Abu Zenima and Hammam Faraun (Fig. 1). This mainly basaltic, Early Tertiary igneous phase was generally interpreted as being related to initial faulting in the Suez rift. Basaltic bodies from northern Sinai have yielded K–Ar ages of 20.2–21.6 Myr (Siedner 1973) i.e. earliest Miocene. Similar ages were obtained for the associated rocks in Egypt (Meneisy & Kreuzer 1974). An age of 24.8 Myr was recorded by Steinitz, Bartov & Hunziker (1978) for the rocks at Raqabat e-Na'ama in central Sinai. Also, the basaltic rocks in west Sinai at Little Gebel Araba, Wadi Nukhul and Wadi Tayiba yielded ages of 24 ± 1.1, 22.3 ± 1.3 and 21.4 ± 1.1 Myr respectively using the K–Ar method for the whole rock (Steen 1982). All information in the area shows that the Suez rift was not active before late Eocene times, about 40 Myr ago, in agreement with Robson (1971), Lowell & Genik (1972), Berggren & Van Couvering (1974) and Hosney (1985). On the other hand, the rift was already well defined and sufficiently depressed to be inundated by the sea in the earliest Miocene (about 20 Myr ago). Thus the early rift-forming events must have occurred between the Late Eocene and the beginning of the Miocene (Garfunkel & Bartov 1977). The palaeomagnetic directions, statistical parameters and pole position results for Wadi Tayiba, Wadi Matulla (Wassif 1988, 1989) and Wadi Nukhul are summarized in Table 5 and Fig. 7. The definition of the ancient magnetization is good for all three basaltic occurrences as shown by the similarity of the mean magnetization directions, the large values of K and the small values of α_{95}. The pole position results for Wadi Tayiba, Wadi Matulla and Wadi Nukhul (Fig. 7) represent new data for the Tertiary period later than 40 Myr. The recently obtained isotopic age determinations (Mossa 1987) yielded K–Ar ages of 22.8 ± 1.4, 23.7 ± 1.9 and 24.8 ± 1.6 Myr for the above-mentioned occurrences respectively, which correspond to the uppermost Oligocene or the earliest Miocene. These pole positions are further contributions to fill the data gap for African poles in the period between 22 and 25 Ma. This Oligo-Miocene volcanic event, exemplified by the previously dated basalts and confirmed by the palaeomagnetic results is related to the rifting of the Gulf of Suez that was well under way at this time (Garfunkel & Bartov 1977). Thus, the spatial relationships between this basaltic magmatism and the Suez rift is evident and in agreement with the conclusion of Harland *et al.* (1982) that opening of the Red Sea began at 24.6 Myr.

It is to be mentioned that there is considerable evidence that the entire Red Sea trough has been formed by the movement of Arabia away from Africa. In addition, a small plate, the Sinai plate is proposed to be separated from the Arabian plate by the Dead Sea fault system (McKenzie & Davies 1970). The Red Sea rifting was preceded in the Early Tertiary by domal uplift of adjacent parts of the Arabian and Nubian parts of the Shield (Brown & Coleman 1972). It is believed that by Late Eocene or Early Oligocene times the Tethys was closed and Africa came into contact with Asia, thus resulting in northerly horizontal compressive force at this location as a result of the collision. The horizontal compressive force will result in two conjugate shear fractures: NW (Suez) and NNE (Aqaba) trends that show relative strength in northern Egypt (Meshref 1982).

ACKNOWLEDGMENTS

I am grateful to the authorities of the Geophysical Lab. of Geological Survey and Mining, where the palaeomagnetic analyses were carried out. I would like to thank Professor Dr G. Saleeb (National Research Centre) and his colleagues for their generous help in the field and in collecting oriented samples.

REFERENCES

Abdalla, A. M., El-Adinani, A. & Fahmi, N., 1963. Stratigraphy of the Lower Mesozoic rocks, western side of Gulf of Suez, Egypt, *Paper No. 27,* UAR Geological Survey.

Andrew, G., 1937. The late Tertiary igneous rocks of Egypt, (field relations), *Bull. Fac. Sci. Cairo Univ.*, **10,** 1–61.

Ball, J., 1916. *The Geography and Geology of West Central Sinai, Egypt,* Survey Department, Cairo.

Basta, E. Z., 1970. Different types of ilmenite–magnetite intergrowths and their origin, *Bull. Fac. Sci. Cairo Univ.*, **44,** 1–195.

Barron, T., 1907. *The Topography and Geology of the Peninsula of Sinai (Western Portion),* Egyptian Survey Department, Cairo.

Barthoux, J. C., 1922. Chronologie et description des roches ignees du Desert Arabique, *Mem Instit. Egypte,* **5,** 1–262.

Berggren, W. A. & Van Couvering, J. A., 1974. The late Neogene, *Paleogeog. Paleoclimat. Paleocol.*, **16,** 1–216.

Brown, G. F. & Coleman, R. G., 1972. The tectonic frame-work of the Arabian Peninsula, *Proc. 24th International Geol. Congr. Montreal,* Section 3, pp. 300–305.

Downloaded from https://academic.oup.com/gji/article/104/2/319/570616 by Curtin University Library user on 06 May 2026

Dietz, R. S. & Holden, T. C., 1974. Reconstruction of Pangea: Breakup and dispersion of continents, Permian to Present, *J. geophys. Res.*, **75,** 4939–4956.

Dunlop, D. J. & Prevot, M., 1982. Magnetic properties and opaque mineralogy of drilled submarine intrusive rocks, *Geophys, J. R. astr. Soc.*, **69,** 763–802.

El-Shazly, E. M., Abdel Hady, M. A., El-Ghawaby, M. A., El-Kassas, I. A. & El-Shazly, M. M., 1974a. Geology of Sinai Peninsula from ERTS-1 Satellite Images, *Remote Sensing Research Project,* Acad. of Sci. Res. & Tec., Cairo.

Embleton, B. J. J. & McElhinny, M. W., 1975. The palaeoposition of Madagascar: Palaeomagnetic evidence from the Isalo Group, *Earth planet. Sci. Lett.*, **27,** 329–341.

Fisher, R. A., 1953. Dispersion on a sphere, *Proc. R. Soc. Lond.*, A, **217,** 295–305.

Garfunkel, Z. & Bartov, Y., 1977. The tectonics of the Suez rift, *Geol. Surv. Israel Bull.*, **71,** 1–44.

Harland, W. B., Cox, A. V., Uewellyn, P. G., Picton, C. A. G., Smith, A. G. & Walters, R., 1982. *A Geologic Time Scale,* Cambridge Earth Science series, Cambridge University Press, Cambridge, UK.

Hosney, H. M., 1985. *Geophysical Investigations in the Gulf of Suez—consequence for the Tectonic Evolution of the Northern Red Sea,* Mathematisch-Naturwissenschaftlichen Fakultat, Christian-Albrechts-Universitat, Kiel.

Hume, W. F., Madgwick, T. G., Moon, F. W. & Sadek, H., 1920. Preliminary geological report of the occurrences of petroleum in Western Sinai, *Petrol. Res. Bull. 2,* Government Press, Cairo.

Irving, E. & Irving, A., 1982. Apparent polar wander paths Carboniferous through Cenozoic and assembly of Gondwana, *Geophys. Surv.*, **5,** 141–188.

Koenigsberger, J. G., 1938. Natural residual magnetism of eruptive rocks, *Terr. Magn. Atmos. Elect.*, **43,** 119–130.

Lowell, J. D. & Genik, G. J., 1972. Sea floor spreading and structural evolution of the southern Red Sea, *Am. Assoc. Petrol. Geol. Bull.*, **56,** 247–259.

McElhinny, M. W. & Embleton, B. J. J., 1976. The palaeoposition of Madagascar: Remanence and magnetic properties of Late Palaeozoic sediments, *Earth planet. Sci. Lett.*, **31,** 101–12.

McKenzie, D. P. & Davies, D., 1970. Plate tectonics of the Red Sea and East Africa, *Nature,* **226,** 1–6.

Meneisy, M. Y. & Kreuzer, H., 1974. Potassium–Argon ages of Egyptian basaltic rocks, *Geol. Jb. Hannover,* **9,** 21–31.

Meshref, W. M., 1982. Regional structural setting of northern Egypt, *Sixth EGPC Exploration Seminar,* Gupco.

Mossa, H. E., 1987. Geologic studies and genetic correlation of basaltic rocks in West Central Sinai, *PhD thesis,* Fac. Sci., Ain Shams University.

Ozima, M. & Larson, E. E., 1970. Low- and high-temperature oxidation of titanomagnetite in relation to irreversible changes in the magnetic properties of submarine basalts, *J. geophys. Res.*, **15,** 1003–1018.

Robson, D. A., 1971. The structure of the Gulf of Suez (clysmic) Rift, with special reference to the eastern side, *Geol. Soc. London Q. J.*, **115,** 247–276.

Said, R., 1962. *The Geology of Egypt,* Elsevier, Amsterdam.

Siedner, G., 1973. K–Ar chronology of Cenozoic volcanics from Northern Israel and Sinai, *Fortschritte der Mineralogie,* **50,** 129–130.

Steen, G., 1982. Radiometric age dating and tectonic significance of some Gulf of Suez igneous rocks, *GUPCO Report ER-82-2.*

Steinitz, G., Bartov, Y. & Hunziker, J. C., 1978. K–Ar age determinations of some Miocene–Pliocene basalts in Israel: their significance to the tectonics of the Rift Valley, *Geol. Mag.*, **115,** 329.

Tarling, D. H., 1980. Models for the fragmentation of Gondwanaland, in *Gondwana Five,* pp. 261–266, eds Cresswell, M. M. & Vella, P., Balkema, Rotterdam.

Tarling, D. H. & Kent, P. E., 1976. The Madagascar controversy still lives, *Nature,* **261,** 304–305.

Wassif, N. A., 1988. Data concerning the magnetic, palaeomagnetic and mineralogical properties of some basalts from Wadi Matulla and Wadi Budra, Sinai, Egypt, *Ain Shams Sci. Bull.*, **27B,** 13–35.

Wassif, N. A., 1989. Some magnetic and mineralogical aspects of Wadi Tayiba Basalts, West Central Sinai, Egypt, *Delta J. Sci. Tanta Univ.*, **13(1),** 247–268.

Weissbrod, T., 1969. The Paleozoic of Israel and adjacent countries, *Geol. Surv. Isr. Bull. No. 47,* parts 1 and 2.

Wilson, R. L., 1959. Remanent magnetism of late Secondary and early Tertiary British rocks, *Phil. Mag.*, **4,** 750–755.

Zijderveld, J. D. A., 1967. A. C. demagnetization of rocks: analysis of results, in *Methods in Paleomagnetism,* eds Collinson, D. W., Creer, K. M. & Runcorn, S. K., Elsevier, Amsterdam.

Downloaded from https://academic.oup.com/gji/article/104/2/319/570616 by Curtin University Library user on 06 May 2026